

In-Depth Docker Commands + Examples (Cheat Sheet)

IMAGE COMMANDS

List Images

`docker images`

Example: Lists all downloaded images.

Pull Image

`docker pull nginx`

Example: Downloads the official NGINX image.

Build Image

`docker build -t myapp:1.0 .`

Example: Builds image from Dockerfile in current directory.

Remove Image

`docker rmi myapp:1.0`

Example: Deletes the specified image.

Tag Image

`docker tag myapp:1.0 myrepo/myapp:latest`

Example: Prepares image for pushing.

Push Image

`docker push myrepo/myapp:latest`

Example: Uploads image to registry.

CONTAINER COMMANDS

Run Container

`docker run nginx`

Example: Runs default NGINX container.

Run in Background

`docker run -d nginx`

Example: Starts container in detached mode.

Port Mapping

`docker run -p 8080:80 nginx`

Example: Exposes port 80 as 8080 on host.

Mount Volume

`docker run -v ./data:/app/data myapp`

Example: Shares local folder into container.

Container Lifecycle

`docker start id`

`docker stop id`

`docker restart id`

Example: Basic container operations.

View Logs

`docker logs myapp`

Example: Shows stdout logs.

Open Shell

`docker exec -it myapp bash`

Example: Interact with a running container.

VOLUMES

List Volumes

docker volume ls

Create Volume

docker volume create appdata

Inspect Volume

docker volume inspect appdata

Delete Volume

docker volume rm appdata

NETWORKS

List Networks

`docker network ls`

Create Network

`docker network create appnet`

Attach Container

`docker network connect appnet myapp`

Inspect Network

`docker network inspect appnet`

SYSTEM CLEANUP

Remove Orphaned Images

`docker image prune`

Remove All Unused

`docker system prune -a`

Disk Usage

`docker system df`

DOCKER COMPOSE

Start Services

`docker compose up -d`

Stop Services

`docker compose down`

Rebuild Containers

`docker compose up --build`

View Logs

`docker compose logs -f`

DOCKERFILE REFERENCES

FROM node:18 — Base image

RUN apt-get install ... — Execute commands

COPY . /app — Copy local files

WORKDIR /app — Set working dir

CMD ["npm","start"] — Default command